

In the format provided by the authors and unedited.

Gate-tunable third-order nonlinear optical response of massless Dirac fermions in graphene

Tao Jiang^{1,9}, Di Huang^{1,9}, Jinluo Cheng², Xiaodong Fan^{3,4}, Zhihong Zhang⁵, Yuwei Shan¹, Yangfan Yi¹, Yunyun Dai¹, Lei Shi^{1,6}, Kaihui Liu⁵, Changgan Zeng^{3,4}, Jian Zi^{1,6}, J. E. Sipe⁷, Yuen-Ron Shen^{1,8}, Wei-Tao Liu^{1,6*} and Shiwei Wu^{1,6*}

¹State Key Laboratory of Surface Physics, Key Laboratory of Micro and Nano Photonic Structures (MOE), Department of Physics, and Institute for Nanoelectronic Devices and Quantum Computing, Fudan University, Shanghai, China. ²Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, Jilin, China. ³International Center for Quantum Design of Functional Materials, Hefei National Laboratory for Physical Sciences at the Microscale, and Synergetic Innovation Center of Quantum Information and Quantum Physics, University of Science and Technology of China, Hefei, China. ⁴CAS Key Laboratory of Strongly-Coupled Quantum Matter Physics, and Department of Physics, University of Science and Technology of China, Hefei, China. ⁵State Key Laboratory for Mesoscopic Physics and School of Physics, Peking University, Beijing, China. ⁶Collaborative Innovation Center of Advanced Microstructures, Nanjing, China. ⁷Department of Physics, University of Toronto, Toronto, Ontario, Canada. ⁸Physics Department, University of California, Berkeley, CA, USA. ⁹These authors contributed equally: Tao Jiang, Di Huang. *e-mail: [wtliu@fudan.edu.cn](mailto:wtlou@fudan.edu.cn); swwu@fudan.edu.cn

In the format provided by the authors and unedited.

Gate-tunable third-order nonlinear optical response of massless Dirac fermions in graphene

Tao Jiang^{1,9}, Di Huang^{1,9}, Jinluo Cheng², Xiaodong Fan^{3,4}, Zhihong Zhang⁵, Yuwei Shan¹, Yangfan Yi¹, Yunyun Dai¹, Lei Shi^{1,6}, Kaihui Liu⁵, Changgan Zeng^{3,4}, Jian Zi^{1,6}, J. E. Sipe⁷, Yuen-Ron Shen^{1,8}, Wei-Tao Liu^{1,6*} and Shiwei Wu^{1,6*}

¹State Key Laboratory of Surface Physics, Key Laboratory of Micro and Nano Photonic Structures (MOE), Department of Physics, and Institute for Nanoelectronic Devices and Quantum Computing, Fudan University, Shanghai, China. ²Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, Jilin, China. ³International Center for Quantum Design of Functional Materials, Hefei National Laboratory for Physical Sciences at the Microscale, and Synergetic Innovation Center of Quantum Information and Quantum Physics, University of Science and Technology of China, Hefei, China. ⁴CAS Key Laboratory of Strongly-Coupled Quantum Matter Physics, and Department of Physics, University of Science and Technology of China, Hefei, China. ⁵State Key Laboratory for Mesoscopic Physics and School of Physics, Peking University, Beijing, China. ⁶Collaborative Innovation Center of Advanced Microstructures, Nanjing, China. ⁷Department of Physics, University of Toronto, Toronto, Ontario, Canada. ⁸Physics Department, University of California, Berkeley, CA, USA. ⁹These authors contributed equally: Tao Jiang, Di Huang. *e-mail: wliu@fudan.edu.cn; swwu@fudan.edu.cn

Supporting Information for

**Gate tunable third-order nonlinear optical response of massless Dirac fermions in
graphene**

Tao Jiang,^{1†} Di Huang,^{1†} Jinluo Cheng,² Xiaodong Fan,^{3,4} Zhihong Zhang,⁵ Yuwei
Shan,¹ Yangfan Yi,¹ Yunyun Dai,¹ Lei Shi,^{1,6} Kaihui Liu,⁵ Changgan Zeng,^{3,4} Jian Zi,^{1,6}
J.E. Sipe,⁷ Yuen-Ron Shen,^{1,8} Wei-Tao Liu,^{1,6*} Shiwei Wu^{1,6*}

† T. Jiang and D. Huang equally contributed to this work.

* Corresponding email: swwu@fudan.edu.cn, wtliu@fudan.edu.cn

The supplementary information includes:

Supplementary Texts 1-5.

Supplementary Table S1.

Supplementary Figures S1-S6.

Supplementary Text

1. Relationship between μ and V_g for ion-gel gated graphene.

The chemical potential μ or Fermi level E_F in graphene could be controlled by adjusting the carrier density n in a field effect transistor (FET). The relationship between μ and n is:

$$\mu(n) = \hbar|v_F|\sqrt{\pi n} \quad (1-1)$$

where v_F is the Fermi velocity of graphene (1×10^6 m/s). $\mu(n)$ can be tuned by the gate voltage (V_g) via¹:

$$V_g - V_{CNP} = \frac{\mu(n)}{e} + \frac{ne}{C_g} \quad (1-2)$$

where V_{CNP} is the voltage at the charge neutral point (CNP) determined from the peak position in the resistance curve, and C_g is the gate capacitance determined from fitting of μ versus V_g in Fig. 1b with Eqs. (1-1) and (1-2). The first term on the right hand side of Eq. (1-2), $\mu(n)/e$, is the quantum capacitance of graphene¹. Individual ion-gel gated graphene devices are characterized by C_g^{1-7} , which varies in the range of 1.4-5.6 $\mu\text{F}/\text{cm}^2$. Thus, only 2-3 Volts of V_g is needed to reach a maximum doping concentration of $6 \times 10^{13} \text{ cm}^{-2}$, corresponding to a chemical potential μ of 0.9 eV⁶.

The chemical potential μ at given V_g was experimentally determined from the infrared (IR) transmission spectrum of graphene. In terms of the surface conductivity of graphene $\sigma^{(1)}$, the normally incident transmission spectrum normalized against that at $V_g = V_{CNP}$ has the expression⁹:

$$T(V_g)/T(V_{\text{CNP}}) = \frac{\left| \frac{1}{\eta_1} + \frac{1}{\eta_2} + \sigma_{\text{V}_{\text{CNP}}} \right|^2}{\left| \frac{1}{\eta_1} + \frac{1}{\eta_2} + \sigma_{V_g} \right|^2}, \quad (1-3)$$

where $\eta_1 = \sqrt{\mu_1/\varepsilon_1}$ and $\eta_2 = \sqrt{\mu_2/\varepsilon_2}$ are the wave impedances of fused silica and ion-gel, respectively. According to Kubo's formula⁸⁻¹⁰, $\sigma^{(1)}$ at $k_B T \ll |\mu|$, $\hbar\omega$ can be approximated by^{9, 10}

$$\sigma^{(1)} \cong \frac{ie^2 k_B T}{\pi \hbar (\hbar\omega + i\Gamma_i)} \left(\frac{\mu}{k_B T} + 2 \ln \left(e^{-\frac{\mu}{k_B T}} + 1 \right) \right) + \frac{ie^2}{4\pi\hbar} \ln \left(\frac{2|\mu| - (\hbar\omega + i\Gamma_e)}{2|\mu| + (\hbar\omega + i\Gamma_e)} \right) \quad (1-4)$$

where Γ_e and Γ_i are the interband and intraband damping factors, respectively. As shown in Fig. S4, fitting the spectra taken at different V_g with Eqs.(1-3) and (1-4) allows the deduction of $\mu(V_g)$.

2. Nonvanishing $\chi^{(3)}$ elements in graphene.

Graphene belongs to the symmetric point group D_{6h} ($6/mmm$), the nonvanishing $\chi^{(3)}$ elements of which for third-order nonlinear polarization induced in a plane perpendicular to the 6-fold axis are known to be^{11, 12}

$$\chi_{xxxx}^{(3)} = \chi_{yyyy}^{(3)} = \chi_{xxyy}^{(3)} + \chi_{xyxy}^{(3)} + \chi_{xyyx}^{(3)}, \text{ and } \begin{cases} \chi_{xxyy}^{(3)} = \chi_{yyxx}^{(3)} \\ \chi_{xyxy}^{(3)} = \chi_{yxyx}^{(3)} \\ \chi_{xyyx}^{(3)} = \chi_{yxxy}^{(3)} \end{cases} \quad (2-1)$$

where x and y are two orthogonal axes in the plane. Thus for third-order nonlinear processes in graphene, if the inputs are all linearly polarized along a certain direction in the plane, the output must be linearly polarized along the same direction and is azimuthally isotropic.

3. Microscopic expressions of $\chi^{(3)}$.

We reproduce here the microscopic expressions of the nonvanishing $\chi^{(3)}$ elements of graphene derived by Cheng *et al.*¹³ using the linearly dispersed two-band model. For four-wave mixing of $\omega_t = \omega_1 + \omega_2 + \omega_3$ in the absence of resonant damping at zero temperature, they are

$$\begin{aligned} \begin{bmatrix} \chi_{xxxy}^{(3)} \\ \chi_{xyxy}^{(3)} \\ \chi_{xyyx}^{(3)} \end{bmatrix} = & -\frac{(\hbar v_F e^2)^2}{4\pi\hbar^5 \varepsilon_0 d_{gr} \omega_t} \left\{ \frac{G\left(\frac{\hbar\omega_1}{2|\mu|}\right)}{6(\omega_1+\omega_2)(\omega_2+\omega_3)(\omega_3+\omega_1)} \left[-\frac{\omega_t(\omega_1\omega_2+\omega_2\omega_3+\omega_3\omega_1)}{\omega_2^2\omega_3^2} A^{(1)} + \right. \right. \\ & \frac{(\omega_1+\omega_2)\omega_t-\omega_1\omega_2}{\omega_2^2\omega_t} A^{(2)} + \frac{(\omega_1+\omega_3)\omega_t-\omega_1\omega_3}{\omega_3^2\omega_t} A^{(3)} \left. \right] + \frac{(\omega_1+\omega_2)G\left(\frac{\hbar\omega_1+\hbar\omega_2}{2|\mu|}\right)}{6\omega_1^2\omega_2^2\omega_3^2\omega_t} [(\omega_1\omega_t - \omega_2\omega_3)A^{(1)} + \\ & (\omega_2\omega_t - \omega_1\omega_3)A^{(2)}] - \frac{\omega_t G\left(\frac{\hbar\omega_t}{2|\mu|}\right)}{6(\omega_1+\omega_2)(\omega_2+\omega_3)(\omega_3+\omega_1)\omega_2^2\omega_3^2} [(\omega_2 + \omega_3)\omega_t - \omega_2\omega_3]A^{(1)} + \\ & c.p. \left. \right\}. \end{aligned} \quad (3-1)$$

where c.p. stands for the simultaneous cycle permutation of indices in $\{\omega_1, \omega_2, \omega_3\}$ and $\{A^{(1)}, A^{(2)}, A^{(3)}\}$ with

$$A^{(1)} = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}, A^{(2)} = \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}, A^{(3)} = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}.$$

and $G(x) = \ln \left| \frac{1+x}{1-x} \right| + i\pi H(|x| - 1)$, with H denoting the Heaviside function. At $\mu = 0$, Eq. (3-1) yields

$$\chi_{xxxx}^{(3)} = \chi_{xxxy}^{(3)} + \chi_{xyxy}^{(3)} + \chi_{xyyx}^{(3)} = \frac{i(\hbar v_F e^2)^2 / (4\hbar^5 \varepsilon_0 d_{gr})}{(\omega_1+\omega_2)(\omega_2+\omega_3)(\omega_3+\omega_1)\omega_t^2} \quad (3-2)$$

For THG with $\omega_1 = \omega_2 = \omega_3 = \omega_0$ and $\omega_t = 3\omega_0$, we have

$$\begin{aligned}\chi_{xxxx}^{(3)} &= 3\chi_{xxyy}^{(3)} = 3\chi_{xyxy}^{(3)} = 3\chi_{xyyx}^{(3)} \\ &= -\frac{(\hbar v_F e^2)^2}{576\pi\epsilon_0 d_{gr}(\hbar\omega_0)^5} \left[17G\left(\frac{\hbar\omega_0}{2|\mu|}\right) - 64G\left(\frac{2\hbar\omega_0}{2|\mu|}\right) + 45G\left(\frac{3\hbar\omega_0}{2|\mu|}\right) \right].\end{aligned}\quad (3-3)$$

At $\mu = 0$, we find $\chi_{xxxx}^{(3)} = \frac{2(\hbar v_F e^2)^2}{576\pi\epsilon_0 d_{gr}(\hbar\omega_0)^5}$ and $\chi_{xxxx}^{(3)} = 1.0 \times 10^{-19} \text{ m}^2 \text{V}^{-2}$ if $\hbar\omega_0 = 0.956 \text{ eV}$.

For SFM of $\omega_t = \omega_1 + 2\omega_2$, we have

$$\begin{aligned}\chi_{xxxx}^{(3)}(\omega_1 + 2\omega_2; \omega_1, \omega_2, \omega_2) &= \\ &= -\frac{(\hbar v_F e^2)^2}{4\pi\hbar^5\epsilon_0 d_{gr}(\omega_1+2\omega_2)} \left[\frac{\omega_1(2\omega_1^2+7\omega_1\omega_2+8\omega_2^2)}{12\omega_2^4(\omega_1+\omega_2)^2(\omega_1+2\omega_2)} G\left(\frac{\hbar\omega_1}{2|\mu|}\right) + \right. \\ &\quad \frac{10\omega_1^2+20\omega_1\omega_2+4\omega_2^2}{12\omega_1^2\omega_2(\omega_1+\omega_2)^2(\omega_1+2\omega_2)} G\left(\frac{\hbar\omega_2}{2|\mu|}\right) - \frac{(\omega_1+\omega_2)^3}{3\omega_1^2\omega_2^4(\omega_1+2\omega_2)} G\left(\frac{\hbar(\omega_1+\omega_2)}{2|\mu|}\right) - \\ &\quad \left. \frac{4\omega_2^3}{3\omega_1^2\omega_2^4(\omega_1+2\omega_2)} G\left(\frac{\hbar(2\omega_2)}{2|\mu|}\right) + \frac{(\omega_1+2\omega_2)^2(2\omega_1^2+\omega_1\omega_2+2\omega_2^2)}{12\omega_1^2\omega_2^4(\omega_1+\omega_2)^2} G\left(\frac{\hbar(\omega_1+2\omega_2)}{2|\mu|}\right) \right] \quad (3-4)\end{aligned}$$

and for DFM of $\omega_t = -\omega_2 + 2\omega_1$,

$$\begin{aligned}\chi_{xxxx}^{(3)}(-\omega_2 + 2\omega_1; -\omega_2, \omega_1, \omega_1) &= \\ &= -\frac{(\hbar v_F e^2)^2}{4\pi\hbar^5\epsilon_0 d_{gr}(-\omega_2+2\omega_1)} \left[\frac{-\omega_2(2\omega_2^2-7\omega_2\omega_1+8\omega_1^2)}{12\omega_1^4(-\omega_2+\omega_1)^2(-\omega_2+2\omega_1)} G\left(\frac{-\hbar\omega_2}{2|\mu|}\right) + \right. \\ &\quad \frac{10\omega_2^2-20\omega_2\omega_1+4\omega_1^2}{12\omega_2^2\omega_1(-\omega_2+\omega_1)^2(-\omega_2+2\omega_1)} G\left(\frac{\hbar\omega_1}{2|\mu|}\right) - \frac{(-\omega_2+\omega_1)^3}{3\omega_1^2\omega_2^4(-\omega_2+2\omega_1)} G\left(\frac{\hbar(-\omega_2+\omega_1)}{2|\mu|}\right) - \\ &\quad \left. \frac{4\omega_1^3}{3\omega_2^2\omega_1^4(-\omega_2+2\omega_1)} G\left(\frac{\hbar(2\omega_1)}{2|\mu|}\right) + \frac{(-\omega_2+2\omega_1)^2(2\omega_2^2-\omega_1\omega_2+2\omega_1^2)}{12\omega_2^2\omega_1^4(-\omega_2+\omega_1)^2} G\left(\frac{\hbar(-\omega_2+2\omega_1)}{2|\mu|}\right) \right] \quad (3-5)\end{aligned}$$

When $\omega_1 - \omega_2 = \Delta\omega \ll \omega_1, \omega_2$, Eq. (3-5) can be approximated by:

$$\begin{aligned}\chi_{xxxx}^{(3)}(-\omega_2 + 2\omega_1; -\omega_2, \omega_1, \omega_1) &= -\frac{(\hbar v_F e^2)^2}{4\pi\hbar^5\epsilon_0 d_{gr}(-\omega_2+2\omega_1)^2\omega_1^3} \left[-\frac{1}{3}\left(\frac{\Delta\omega}{\omega_1}\right)^3 G\left(\frac{\hbar(\Delta\omega)}{2|\mu|}\right) - \right. \\ &\quad \left. \frac{1}{4}\left(\frac{\omega_1}{\Delta\omega}\right)^2 G\left(\frac{-\hbar\omega_2}{2|\mu|}\right) - \frac{1}{2}\left(\frac{\omega_1}{\Delta\omega}\right)^2 G\left(\frac{\hbar\omega_1}{2|\mu|}\right) + \frac{1}{4}\left(\frac{\omega_1}{\Delta\omega}\right)^2 G\left(\frac{\hbar(-\omega_2+2\omega_1)}{2|\mu|}\right) - \frac{4}{3}G\left(\frac{\hbar(2\omega_1)}{2|\mu|}\right) \right] \quad (3-6)\end{aligned}$$

It is seen that the three terms proportional to $(\Delta\omega)^{-2}$ dominate in Eq. (3-6) and diverge when ω_2 approaches ω_1 , as illustrated in Fig. 5e. If $2|\mu| \ll \hbar\omega_1, \hbar\omega_2$, Eq. (3-6) reduces to:

$$\chi_{xxxx}^{(3)} = \frac{i(\hbar v_F e^2)^2 / (4\hbar^5 \varepsilon_0 d_{gr})}{2\omega_1(\omega_1 - \omega_2)^2 (2\omega_1 - \omega_2)^2} \quad (3-7)$$

4. Inclusion of electron temperature and resonant damping effects in simulating $|\chi^{(3)}|$.

Damping of resonant transitions and finite electron temperature have been incorporated into Eq.(3-1) by Cheng *et al.*¹⁴, leading to suppression of singularities and smearing and broadening of sharp features. We assumed two phenomenological resonant damping factors, Γ_e for interband transitions and Γ_i for intraband transitions, in our simulation. Unless specified otherwise, we used $\Gamma_e = 0.2 \times |\mu|$ eV and $\Gamma_i = 0.5$ meV in comparing simulation with experiment. (Simulations with different values of damping factors are shown in Fig. S5.). Photo-excited carriers in graphene can establish quasi-thermal equilibrium within a few tens of femtoseconds¹⁵. The resulting electron temperature T_e can be much higher than the room temperature (300 K), if the excited carrier density is high. Generally, T_e depends on the incident fluence F , and the chemical potential μ . When one-photon interband transitions are allowed for $2|\mu| < \hbar\omega_1$ (assuming $\omega_1 > \omega_2$), T_e can reach as high as a few thousand degrees^{15, 16}. When one-photon transitions are blocked ($2|\mu| > \hbar\omega_1$), electron excitations to the

conduction band are weak and the electron temperature remains practically at the room temperature (300 K).

We obtained T_e as a function of μ for a given input fluence as follows. For $\mu = 0$, we can follow Refs. 15 and 16 to estimate $T_e(\mu = 0)$ versus input fluence (with a pulse width of 200 fs in our case), which is plotted in Fig. S6a. Corresponding to an incident laser fluence F of 3.14 J/m^2 at 1300 nm, the estimated $T_e(\mu = 0)$ is about 2100 K. For $2|\mu| \gg \hbar\omega_0$ ($\sim 0.96 \text{ eV}$), we have $T_e = T_0 \sim 300 \text{ K}$. In between $2|\mu| \gg \hbar\omega_0$ and $\mu = 0$, we expect a smooth variation from T_0 to $T_e(\mu = 0)$ over a range of about 92 meV around $2|\mu| = \hbar\omega_0$, which comes from fitting of the linear absorbance measurements in Fig. S4. The approximate T_e versus $|\mu|$ is depicted in Fig. S6b. Knowing T_e versus $|\mu|$ for given input fluence, we can then follow Ref. 14 to calculate $|\chi^{(3)}|$ versus μ .

As seen in Fig. 4c and Fig. S1, the effect of significantly higher electron temperature than 300 K for $2|\mu| \leq \hbar\omega_0$ on THG is not very significant because $|\chi^{(3)}|$ in this range is small. The same is true for SFM (Fig. 4b). The effect on DFM is more appreciable (Fig. 5d and Fig. S2); as $|\mu|$ decrease towards zero, the rise of $|\chi^{(3)}|$ is less and more spread out.

5. Estimation of $|\chi^{(3)}|$ for THG and FWM signals.

The third-order nonlinear optical polarization sheet induced in a graphene monolayer of thickness d_{gr} is given by

$$P^{(3)}(\omega_t = \omega_1 + \omega_2 + \omega_3) = \chi^{(3)} d_{gr} E(\omega_1) E(\omega_2) E(\omega_3) \quad (5-1)$$

and the output field generated by the polarization sheet is^{11,12}

$$E(\omega) = \alpha \frac{1}{4} \frac{i\omega}{2n_\omega c} P^{(3)}(\omega) \quad (5-2)$$

where $\alpha = 1$ for THG and $\alpha = 3$ for two-colour FWM.

The relation between the field and the corresponding average power is

$$P(\omega_i) = \frac{1}{8} \left(\frac{\pi}{\ln 2} \right)^{3/2} f \tau W^2 n_\omega \epsilon_0 c \frac{|E(\omega)|^2}{2} \quad (5-3)$$

assuming pulses with repetition rate f , pulse duration τ , and a Gaussian spatial profile of diameter W on the sample.

In our experiment of THG from graphene, we had $d_{gr} = 0.33$ nm, $f = 80$ MHz, $\tau \sim 200$ fs, $W \sim 1.5$ μ m, and the refractive indices taken as $n_{\omega_0} = n_{3\omega_0} = 2.4$. The typical input power at 1300 nm ($\hbar\omega_0 = 0.956$ eV) is 0.68 mW, and $E(\omega_0)$ was estimated to be $\sim 7 \times 10^5$ V/cm. The THG signal was detected in the photon counting mode, and calibrated by a beam of known power at $3\omega_0$ propagating along the same path to the same detector¹⁷. We obtained $|\chi_{xxxx}^{(3)}| = 1.9 \pm 0.3 \times 10^{-19} \text{ m}^2/\text{V}^2$ at $\mu \sim 0$, in fair agreement with the theoretical value of $1.0 \times 10^{-19} \text{ m}^2/\text{V}^2$.

Similarly, $\chi^{(3)}$ for both sum-frequency mixing (SFM) and difference-frequency mixing (DFM) in graphene could be estimated¹¹.

Supplementary Table

Third-order nonlinear optical processes	References	$ \chi^{(3)} $ (m ² /V ²)
FWM	PRL 105, 097401 (2010) ¹⁸	10 ⁻¹⁵
	Nano Lett. 15, 4968 (2015) ¹⁹	10 ⁻¹⁴
THG	PRB 87, 121406 (2013) ¹⁷	10 ⁻¹⁶
	PRX 3, 021014 (2013) ²⁰	10 ⁻¹⁹
	ACS Nano 7, 8441 (2013) ²¹	10 ⁻¹⁵
	2D Materials 4, 011006 (2016) ²²	10 ⁻¹⁹
Optical Kerr effect, two-photon absorption and self-phase modulation	Nat. Photon. 6, 554 (2012) ²³	10 ⁻¹⁷
	Nano Lett. 11, 2622 (2011) ²⁴	10 ⁻¹⁵
	Opt. Lett. 37, 1856 (2012) ²⁵	10 ⁻¹³
	Nano Lett. 11, 5159 (2011) ²⁶	10 ⁻¹⁵
	Phys. Rev. Appl. 6, 044006 (2016) ²⁷	10 ⁻¹⁵
	Opt. Lett. 41, 3281 (2016) ²⁸	10 ⁻¹⁵

Table S1. Measured values of $|\chi^{(3)}|$ of graphene reported in the literature by various groups spanning over a wide range.

Supplementary Figures

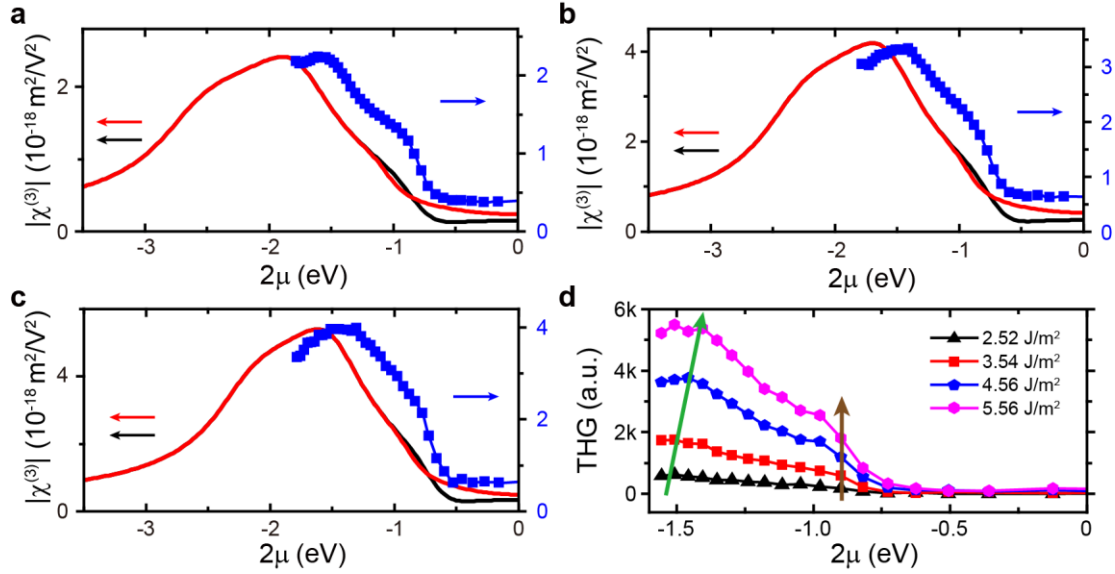


Figure S1. Comparison of theory and experiment of μ -dependent $\chi^{(3)}$ for THG at three different input wavelengths: a-c, 1400 nm, 1566 nm and 1650 nm, respectively.

The theoretical curves, taking into account the heated electron temperature and resonant damping effects, are in red, and those with the electron temperature fixed at 300 K are in black. As references, the estimated heated electron temperature T_e at $\mu = 0$ is 2200 K for **a**, 2030 K for **b**, and 1650 K for **c**, respectively. Interband and intraband damping constants are chosen as $\Gamma_e = 0.2 \times |\mu|$ eV, and $\Gamma_i = 0.5 \text{ meV}^{14}$. The experimental data are taken from Fig. 2c and shown as blue squares. **d**, THG intensity as a function of 2μ under incident laser fluences (at 1380 nm with 200 fs pulse width) of 2.52 J/m^2 (black), 3.54 J/m^2 (red), 4.56 J/m^2 (blue), and 5.56 J/m^2 (magenta), respectively. Two solid arrows (brown and green) mark the apparent shift of the shoulder features. This intensity-dependent downshift is not captured by our perturbation-based theoretical calculations, and may require further study in the future.

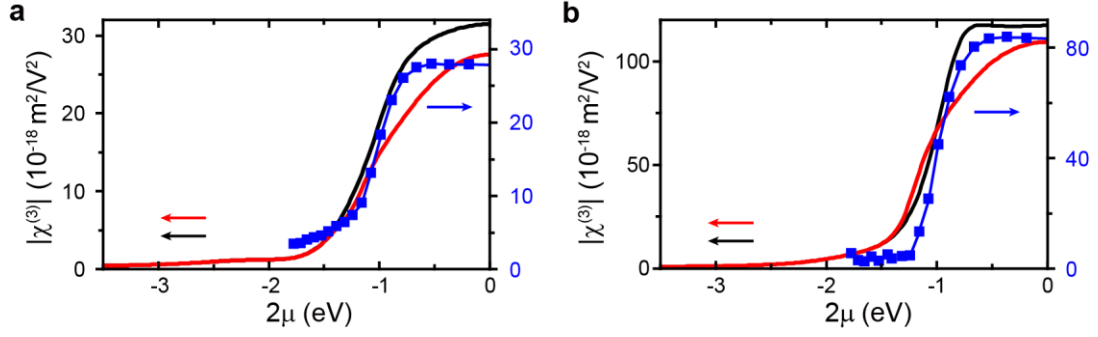


Figure S2. Theoretical calculations of μ -dependent $\chi^{(3)}$ and comparison with experiment for DFM. a, b, Calculated $\chi_{xxxx}^{(3)}$ of DFM ($2\omega_1 - \omega_2$) and DFM ($2\omega_2 - \omega_1$), respectively, with $\hbar\omega_1 = 1.195 \text{ eV}$ and $\hbar\omega_2 = 0.994 \text{ eV}$. The theoretical curves, taking into account the heated electron temperature ($T_e = 1750 \text{ K}$ at $\mu = 0$) and resonant damping effects, are in red, and those with the electron temperature fixed at 300K are in black. Interband and intraband damping constants are chosen as $\Gamma_e = 0.2 \times |\mu| \text{ eV}$, and $\Gamma_i = 0.5 \text{ meV}^{14}$. The experimental data taken from Fig. 3f are shown as blue squares.

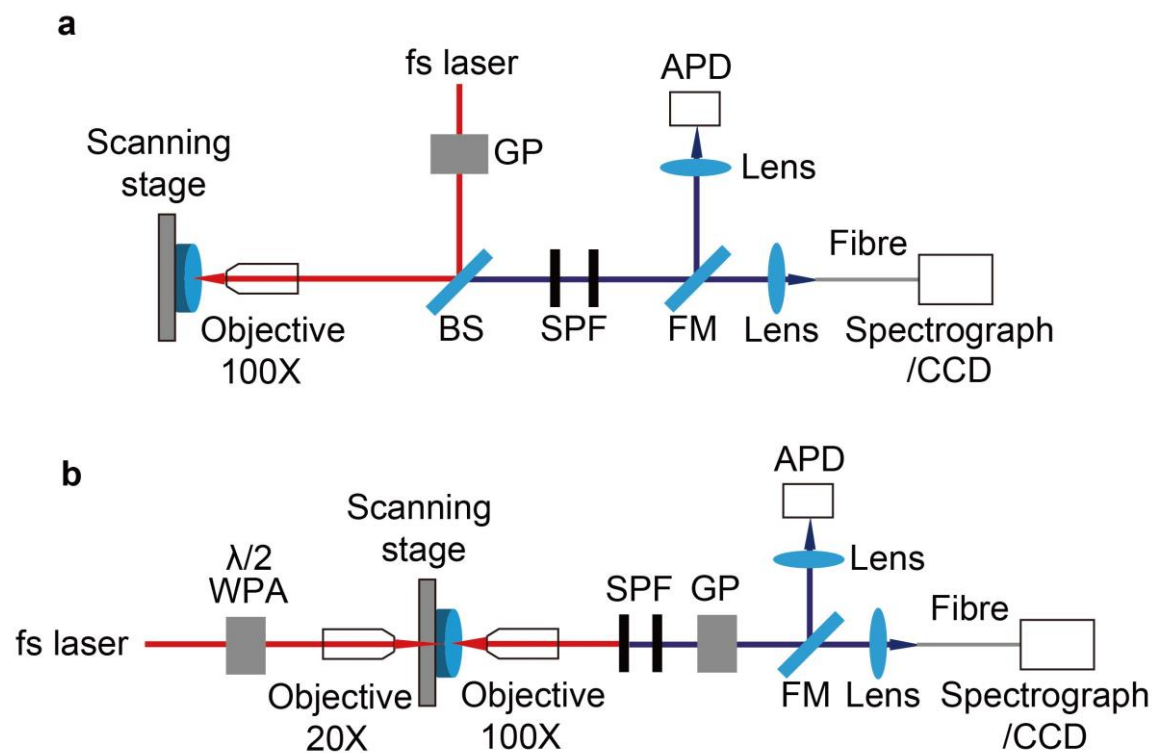


Figure S3. Optical microscopic setups for THG and FWM measurements. a, Output in reflected direction. **b,** Output in transmitted direction. GP: Glan-Thompson polarizer, BS: beamsplitter, SPF: short-pass filters, FM: flip mirror, APD: avalanche photodiode.

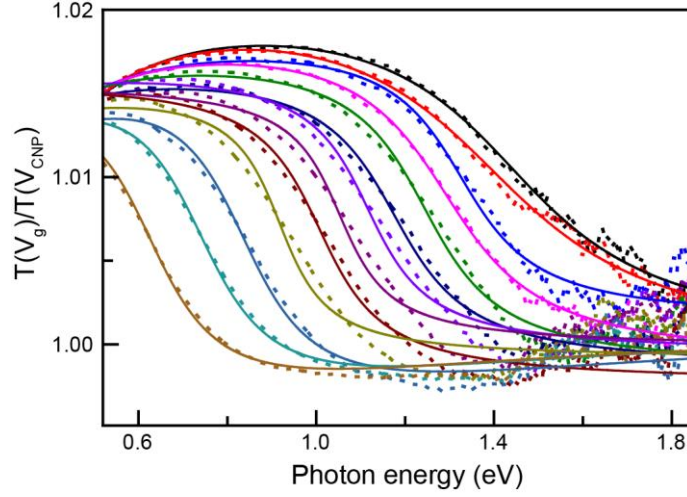


Figure S4. Fitting of the normalized transmittance spectra. The experimental transmittance spectra in Fig. 1c are shown here as dotted curves. From left to right, $V_g - V_{CNP}$ changes from -0.8 to -3.2 V in steps of 0.2 V. The spectra were fitted as solid curves by using the equations (1-3) and (1-4) described in the SI, allowing deduction of chemical potential as a function of V_g , which is shown in Fig. 1b. A small intensity offset to the normalized transmittance was applied in the fitting.

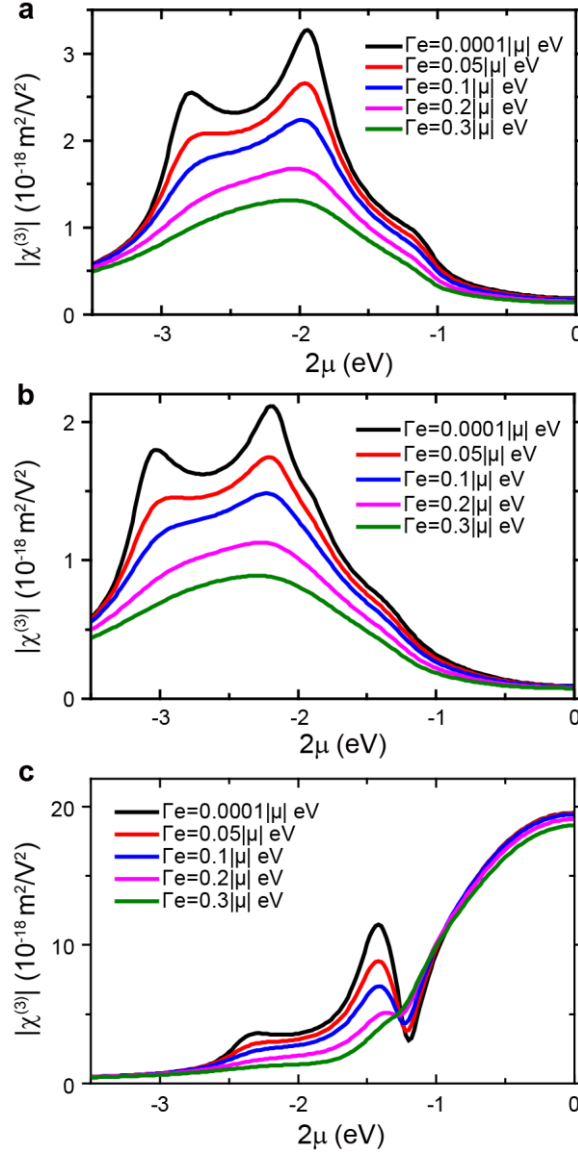


Figure S5. Theoretical calculations of μ -dependent $\chi^{(3)}$ with different damping coefficients Γ_e . **a**, $\chi_{xxxx}^{(3)}$ versus μ for THG with $\hbar\omega_0 = 0.956$ eV calculated at heated electron temperature. **b**, **c**, $\chi_{xxxx}^{(3)}$ versus μ for $(\omega_1 + 2\omega_2)$ SFM and $(2\omega_1 - \omega_2)$ DFM with $\hbar\omega_1 = 1.195$ eV and $\hbar\omega_2 = 0.956$ eV calculated at heated electron temperature, respectively. $\hbar\omega_1 = 1.195$ eV and $\hbar\omega_2 = 0.956$ eV. The heated electron temperature T_e at $\mu = 0$ is estimated to be 2100 K for **a**, 1550 K for **b**, and 1750 K for **c**, respectively, corresponding to an incident fluence of 3.14 J/m^2 for **a**, 0.58 J/m^2 (at ω_l)

and 0.72 J/m^2 (at ω_2) for **b**, and 0.74 J/m^2 (at ω_1) and 1.1 J/m^2 (at ω_2) for **c**. In the calculations, the intraband damping Γ_i is fixed at 0.5 meV, but the interband damping Γ_e is chosen to be $0.0001 \times |\mu| \text{ eV}$, $0.05 \times |\mu| \text{ eV}$, $0.1 \times |\mu| \text{ eV}$, $0.2 \times |\mu| \text{ eV}$, and $0.3 \times |\mu| \text{ eV}$ for **a**, **b**, and **c**, respectively. As expected, smaller Γ_e leads to more prominent resonant features.

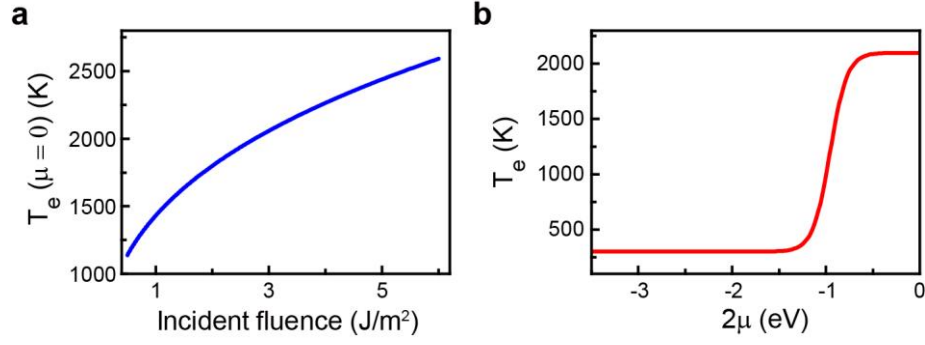


Figure S6. Electron temperature used for simulating $|\chi^{(3)}|$. **a**, Electron temperature T_e at $\mu = 0$ as a function of the incident laser fluence estimated by following Ref. 15 and 16. **b**, Quasi-equilibrium electron temperature T_e as a function of the chemical potential in graphene resulting from electron excitations by an incident pulse of wavelength at 1300 nm, pulse width of 200 fs, and fluence of 3.14 J/m^2 .

References:

1. Das, A. *et al.* Monitoring dopants by Raman scattering in an electrochemically top-gated graphene transistor. *Nat. Nanotech.* **3**, 210-215 (2008).
2. Lu, C., Fu, Q., Huang, S. & Liu, J. Polymer Electrolyte-Gated Carbon Nanotube Field-Effect Transistor. *Nano Lett.* **4**, 623-627 (2004).
3. Cho, J. H. *et al.* Printable ion-gel gate dielectrics for low-voltage polymer thin-film transistors on plastic. *Nat. Mater.* **7**, 900-906 (2008).
4. Ye, J. T. *et al.* Liquid-gated interface superconductivity on an atomically flat film. *Nat. Mater.* **9**, 125-128 (2009).
5. Kim, B. J. *et al.* High-performance flexible graphene field effect transistors with ion gel gate dielectrics. *Nano Lett.* **10**, 3464-3466 (2010).
6. Chen, C. *et al.* Controlling inelastic light scattering quantum pathways in graphene. *Nature* **471**, 617-620 (2011).
7. Tielrooij, K. J. *et al.* Electrical control of optical emitter relaxation pathways enabled by graphene. *Nat. Phys.* **11**, 281-287 (2015).
8. Li, Z. Q. *et al.* Dirac charge dynamics in graphene by infrared spectroscopy. *Nat. Phys.* **4**, 532-535 (2008).
9. Hanson, G. W. Dyadic Green's functions and guided surface waves for a surface conductivity model of graphene. *J. Appl. Phys.* **103**, 064302 (2008).
10. Falkovsky, L. A. & Pershoguba, S. S. Optical far-infrared properties of a graphene monolayer and multilayer. *Phys. Rev. B* **76**, 153410 (2007).

11. Boyd, R. W. *Nonlinear Optics* (Academic Press, New York, 2008).
12. Shen, Y. R. *The Principles of Nonlinear Optics* (Wiley-Interscience, New York, 1984).
13. Cheng, J. L., Vermeulen, N. & Sipe, J.E. Third order optical nonlinearity of graphene. *New J. Phys.* **16**, 53014 (2014).
14. Cheng, J. L., Vermeulen, N. & Sipe, J. E. Third-order nonlinearity of graphene: Effects of phenomenological relaxation and finite temperature. *Phys. Rev. B* **91**, 235320 (2015).
15. Lui, C.H., Mak, K.F., Shan, J., Heinz, T.F. Ultrafast photoluminescence from graphene. *Phys. Rev. Lett.* **105**, 127404 (2010).
16. Shi, S.-F., Tang, T.-T., Zeng, B., Ju, L., Zhou, Q., Zettl, A., Wang, F. Controlling graphene ultrafast hot carrier response from metal-like to semiconductor-like by electrostatic gating. *Nano Lett.* **14**, 1578-1582 (2014).
17. Kumar, N. *et al.* Third harmonic generation in graphene and few-layer graphite films. *Phys. Rev. B* **87**, 121406 (2013).
18. Hendry, E., Hale, P. J., Moger, J., Savchenko, A. K. & Mikhailov, S. A. Coherent nonlinear optical response of graphene. *Phys. Rev. Lett.* **105**, 097401 (2010).
19. Ciesielski, R. *et al.* Graphene near-degenerate four-wave mixing for phase characterization of broadband pulses in ultrafast microscopy. *Nano Lett.* **15**, 4968-4972 (2015).
20. Hong, S. *et al.* Optical Third-Harmonic Generation in Graphene. *Phys. Rev. X* **3**,

021014 (2013).

21. S äyn äjoki A. *et al.* Rapid large-area multiphoton microscopy for characterization of graphene. *ACS Nano* **7**, 8441–8446 (2013).

22. Woodward, R. I. *et al.* Characterization of the second- and third-order nonlinear optical susceptibilities of monolayer MoS₂ using multiphoton microscopy. *2D Materials* **4**, 011006 (2017).

23. Gu, T. *et al.* Regenerative oscillation and four-wave mixing in graphene optoelectronics. *Nat. Photon.* **6**, 554-559 (2012).

24. Yang, H. *et al.* Giant two-photon absorption in bilayer graphene. *Nano Lett.* **11**, 2622-2627 (2011).

25. Zhang, H. *et al.* Z-scan measurement of the nonlinear refractive index of graphene. *Opt. Lett.* **37**, 1856–1858 (2012).

26. Wu, R. *et al.* Purely coherent nonlinear optical response in solution dispersions of graphene sheets. *Nano Lett.* **11**, 5159-5164 (2011).

27. Vermeulen, N. *et al.* Negative Kerr nonlinearity of graphene as seen via chirped-pulse-pumped self-phase modulation. *Phys. Rev. Applied* **6**, 044006 (2016).

28. Dremetsika, E. *et al.* Measuring the nonlinear refractive index of graphene using the optical Kerr effect method. *Opt. Lett.* **41**, 3281-3284 (2016).